

- 1 Write the following numbers in order of size, starting with the smallest.

$$\sqrt{0.81}, \quad 0.902, \quad \frac{399}{441}, \quad 0.86^{\frac{2}{3}}$$

Answer _____, _____, _____, _____ [1]

- 2 Write as a single fraction in its simplest form $\frac{7x}{(x-5)^2} + \frac{1}{5-x}$.

Answer _____ [2]

- 3 The capacity of a SD card is 250 gigabytes correct to 2 significant figures.
The size of a picture file is 2.5 megabytes correct to 1 decimal place.

Calculate the largest number of files that can be stored in this SD card.
Give your answer in standard form, correct to 3 significant figures.
(1 gigabyte = 10^9 bytes, 1 megabyte = 10^6 bytes)

Answer _____ [2]

4 Simplify $\frac{5c}{2} \div \frac{20c^2}{d}$.

Answer _____ [2]

- 5 y is directly proportional to x^2 .
If x is increased by 200%, find the percentage increase in y .

Answer _____ % [2]

- 6 A sum of money was divided between Amelia, Brandon and Claire in the ratio 2 : 3 : 4.
If instead, this money had been divided equally between them, Amelia would have received an extra \$20.
What was the total sum of money?

Answer \$ _____ [2]

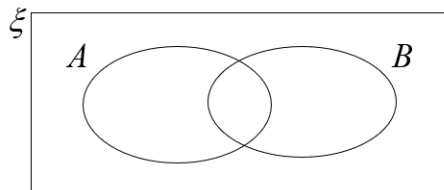
- 7 (a) Solve the inequalities $-8 \leq 2 - 3x < 7 - \frac{1}{2}x$.

Answer _____ [2]

- (b) Write down all the integers that satisfy $-8 \leq 2 - 3x < 7 - \frac{1}{2}x$.

Answer _____ [1]

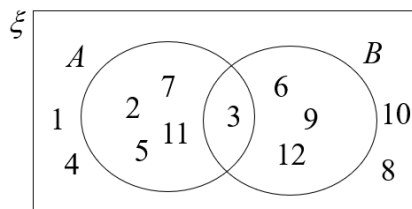
- 8 (a) On the Venn diagram, shade the region which represents $A \cup B'$.



[1]

- (b) $\xi = \{\text{integers } x : 1 \leq x \leq 12\}$
 $A = \{2, 3, 5, 7, 11\}$
 $B = \{3, 6, 9, 12\}$

This information is shown on the Venn diagram.



Use one of the symbols below to complete each statement.

$\emptyset$ $\in$ $\subset$ $\notin$ $\not\subset$

- (i) 7 _____ B [1]

- (ii) $\{3, 7\}$ _____ A [1]